UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

Source: CSC429 (1).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Parallel and multiprocessor architecture

Give suitable example for each processor organization.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A3 (5 marks)

Topic: Data representation and number systems

Calculate -43 - 23 using the sign-and-magnitude method. Assume an 8-bit byte. Show your complete workflow.

Answer: left -43 -> 1 0101011; right 23 -> 0 0010111; -43 - 23 = -66 -> sign-magnitude 1 1000010

- *1 mark for converting both operands to sign-magnitude.*
- *1 mark for applying the sign rule for subtraction.*
- *1 mark for the correct magnitude operation.*
- *1 mark for the correct sign bit.*
- *1 mark for the final decimal and byte representation.*

QUESTION A4 (5 marks)

Topic: Computer generations and history

Question 1 is based on the following image and statements in Figure 1.
Charles Babbage was an English mathematician, a philosopher and also an inventor. He was born on 1791 in London England. He was known for his contributions to invent machine X to solve four basic arithmetic functions. The machine X has improved its calculations design from Stepped Reckoner machine and early mechanical calculator. The machine X has introduced a number of computing concepts which are still in use today. Although this machine was not actually completed, it is the first machine

that deserved to be called as a computer.
Figure 1 : A snippet from biography web page
a) What is the machine X name?
b) In which generation does the machine X introduced?
c) What is the function of machine X?
d) Name the machine that has been invented prior to machine X?
e) What is the title given to Charles Babbage due to his success to invent machine X?

Source: CSC429 (10).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A5 (10 marks)

Topic: Computer architecture and CPU organisation

System interconnection is a mechanism that provides communication among computer components.
a) What is the size of a bus if the word size is 64 bit?
b) List the TWO (2) types of bus.
c) Illustrate the location of a bus between CPU and memory.
d) Differentiate between system interconnection and CPU interconnection.
e) Discuss ONE (1) drawback each for the different types of bus timing.

Source: CSC429 (6).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Data representation and number systems

Answer both parts for the same expression. Assume an 8-bit byte.
Expression: -39 - (-71)

a) Calculate -39 - (-71) using sign and magnitude.

Answer: left -39 -> 1 0100111; right -71 -> 1 1000111; -39 - (-71) = 32 -> sign-magnitude 0 0100000

- *1 mark for operand representations.*
- *1 mark for sign handling.*
- *1 mark for magnitude arithmetic.*
- *1 mark for the result sign bit.*
- *1 mark for the final byte.*

b) Calculate -39 - (-71) using 1's and 2's complement.

Answer: left -39 -> 1's 11011000, 2's 11011001; right -71 -> 1's 10111000, 2's 10111001; -39 - (-71) = 32 -> 1's 00100000, 2's 00100000

- *1 mark for the positive operand in binary.*
- *1 mark for the 1's complement of the negative operand.*
- *1 mark for adding 1 to form the 2's complement.*
- *1 mark for the correct 8-bit result.*
- *1 mark for converting the result back to decimal.*

QUESTION A7 (10 marks)

Topic: Memory hierarchy and cache

Question 7 is based on the following scenarios.
While cleaning his house, Ahmad found an old cassette tape of a popular 80's band. He played the tape and listened to several songs according to the listed tracks.
Hafiz was tasked to research on the history of Malay culture by his professor. He went

to the National Archive to search for information and manage to access one the resources which was stored in a floppy disk.
Sara is planning to buy a new computer. She aims to get one with good processing speed. While researching for computer specifications, she found out that larger cache space has a direct effect towards processing performance.

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Hana loves playing role-playing games (RPG) and is planning to buy a newly released game that requires a high-speed RAM. But, she is devastated when she found out that her game console does not support the minimum RAM requirement of the game.

a)

Based on the above scenarios, identify the access method for each storage devices/memory mentioned in (i), (ii), (iii) and (iv).

b)

Describe SRAM and DRAM.

c) Which level of cache is the fastest in a three-level cache architecture? Which type of RAM is used to implement the cache?

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Input/output and interrupts

a) Why must the CPU be faster than all the other inpuuoutput devices?

b) How do the I/O controllers facilitate communication between the CPU and I/O devices?

c) Discuss TWO (2) types of I/O modules.

d) When would an I/O controller of a specific device be installed?

Source: CSC429 (20).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Computer architecture and CPU organisation

a) Discuss the whole operation of the machine cycle and its respective responsible computer components.

b) Elaborate on what will happen in the machine cycle if the main memory is corrupted.

Source: CSC429 (21).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Input/output and interrupts

Illustration in Figure 1 shows an interrupt that happen during the execution of a program. Question 4 is based on Figure 1.

Update status

Status

The program can't check for definition updates. Error found: Ox80072ee7. The server name or address could not be resolved

Go online to view troubleshpoting tips

Start time

1:03 AM

Time elapse& 00:00:01

Figure 1

Example of interrupt

Define interrupt.

a)

b)

Why interrupt is important?

Explain TWO (2) types of interrupt.

c)

d)

Discuss ONE (1) option to handle multiple interrupts that happen simultaneously.

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 2 and Table 1.

y = A -
2C
* D
B + A2

Figure 2

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three-address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

Table 1 : Machine instruction

Three-address	One-address	Two-address
LOADp		MOV (p,q)
ADD (p.q.r)	STORp	ADD (p,q)
SUB (p,q,r)	ADDp	SUB (p,q)
MUL (p.q.r)	SUBp	MUL (p,q)
DIV(p,q,r)	MULp	DIVp

Source: CSC429 (18).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

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